Name: _____ Date: _____

PS-01: $\varepsilon - \delta$ Proofs, Part I

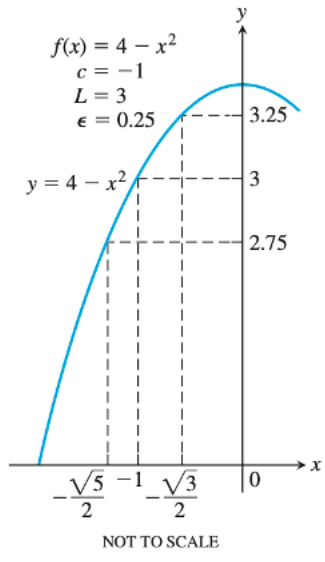
Topics

- Introduction to $\varepsilon - \delta$ definition of a limit
- $\varepsilon - \delta$ limits of linear function

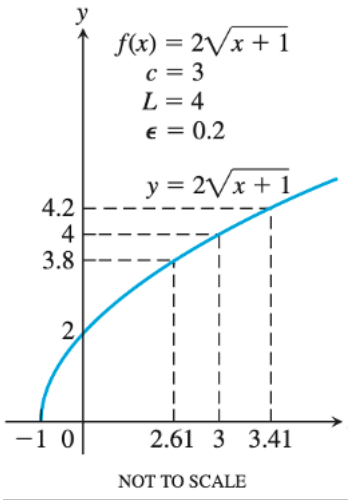
1. In the following exercises, use the graphs to find a $\delta > 0$ such that for all x,

$$0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon$$

(a)



(b)



2. Each of the following exercises gives a function $f(x)$ and numbers L, c, and $\varepsilon > 0$. In each case, find an open interval about c on which the inequality $|f(x) - L| < \varepsilon$ holds. Then, give a value for $\delta > 0$ such that for all x satisfying $0 < |x - c| < \delta$, the inequality $|f(x) - L| < \varepsilon$ holds.

(a)	$f(x) = 2x - 2$	$L = -6$	$c = -2$	$\varepsilon = 0.02$
(b)	$f(x) = \sqrt{x+1}$	$L = 1$	$c = 0$	$\varepsilon = 0.1$

3. Give an $\varepsilon - \delta$ prove of each stated fact

   a) $\lim\limits_{x \to 0} (2x - 1) = -1$

   b) $\lim\limits_{x \to 0} (3x - 3) = -64$

   c) $\lim\limits_{x \to c} (mx + b) = mc + b$

0.1 Solutions

1. This is is from Thomas 2.3.16

$$\begin{aligned} |(2x-2)-(-6)| &< 0.02 \\ |2x+4| &< 0.02 \\ -0.02 < 2x+4 &< 0.02 \\ -4.02 < 2x &< 3.98 \\ -2.01 < x &< 1.99 \end{aligned}$$

3c

Proof. Let $\varepsilon > 0$ be given. We need to find a $\delta > 0$ such that for all x,

$$0 < |x-c| < \delta \implies |f(x)-L| < \varepsilon$$

Let $\varepsilon > 0$ and choose $\delta = \dfrac{\varepsilon}{|m|}$. Then,

$$\begin{aligned} |f(x)-L| &= |(mx+b)-(mc+b)| \\ &= |mx-mc| \\ &= |m| \cdot \underbrace{|x-c|}_{<\delta} \end{aligned}$$

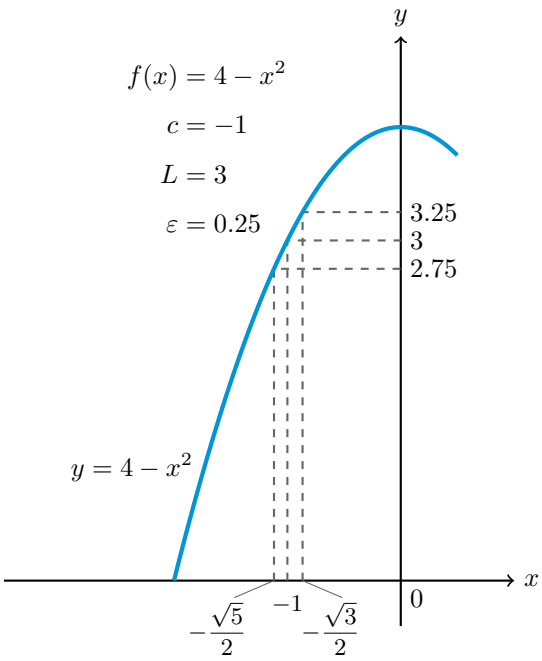
To ensure that $|f(x)-L| < \varepsilon$, we need:

$$|m||x-c| < \varepsilon$$

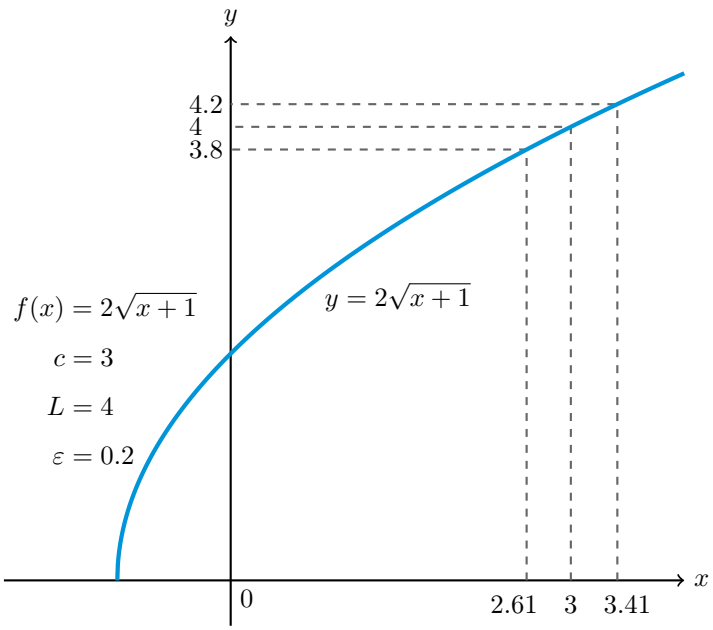
If $m \neq 0$, we can choose $\delta = \dfrac{\varepsilon}{|m|}$. Then, for all x satisfying $0 < |x-c| < \delta$, we have:

$$|f(x)-L| = |m||x-c| < |m| \cdot \frac{\varepsilon}{|m|} = \varepsilon$$

Therefore, $\lim\limits_{x \to c}(mx+b) = mc+b$. □



NOT TO SCALE



NOT TO SCALE

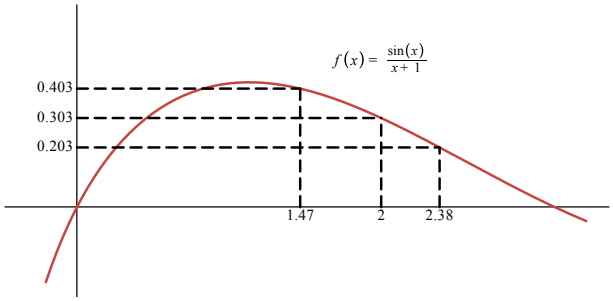


Figure 1: `https://www.desmos.com/calculator/vpiohuxekz`